

dexamine: A Python Package to Examine Decentralized Exchange Data

Magnus Hansson ^{1,2}

¹ Stockholm Business School, Stockholm University ² Swedish House of Finance

DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Open Journals](#) 

Reviewers:

- [@openjournals](#)

Submitted: 01 January 1970

Published: unpublished

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

“dexamine” is a Python package designed to streamline and facilitate the parsing of decentralized exchange (DEX) data on the Ethereum blockchain. Developed in response to the growing complexity and volume of DEX transactions, this tool provides researchers, developers, and analysts with a means to efficiently access, interpret, and analyze Ethereum-based DEX data. The package leverages the fundamental principles outlined in the Ethereum white paper ([Buterin, 2013](#)) and the Ethereum yellow paper ([Wood, 2014](#)), as well as documentation and papers for specific DEXes such as Uniswap ([adams2023?](#)).

By integrating functionalities compatible with Ethereum client implementations like Erigon ([Erigon Team, 2022](#)), and drawing on the data indexing capabilities akin to Trueblocks ([TrueBlocks Team, 2022](#)), dexamine offers a comprehensive solution for DEX data parsing.

Statement of need

The Ethereum blockchain, since its inception, has evolved into a complex ecosystem, especially with the proliferation of DeFi applications. With this evolution comes the challenge of effectively parsing and analyzing the vast and intricate data generated by DeFi protocols. Existing solutions like ([Erigon Team, 2022](#)) provide optimized Ethereum client implementations, while tools like ([trueblocks?](#)) offer efficient blockchain data indexing. Nevertheless, there remains a gap in the form of a dedicated tool specifically designed for parsing DEX data within the Ethereum ecosystem.

The dexamine package addresses this gap by providing a tailored solution that simplifies the complexities involved in parsing DEX transaction data. It is not only a response to the technical needs identified in the DeFi space but also an embodiment of the open-source ethos that Ethereum was built upon. The tool is designed to be user-friendly, adaptable, and efficient, making it an asset for anyone working with decentralized exchange data.

Research project utilizing “dexamine” Mention (if applicable) a representative set of past or ongoing research projects using the software and recent scholarly publications enabled by it.

dexamine example

A basic example of how to parse all market events (swaps, mints, burns) from a Uniswap v3 transaction receipt:

```
# Import packages
from dexamine.parsers.uniswap_v3 import parse_uni_v3_events
from dexamine.shared import general_helpers
from dexamine.shared import constants
```

```
# Test transaction
tx_hash = "0xf16f579a54c0d5700310ca54948d315b65b3f7d224a4af621ebfbaf29fcae8d5"

# Get logs from the transaction
receipt_data = general_helpers.get_receipt_data_by_hash(tx_hash)
logs = receipt_data['logs']

# Load ABIs
erc20_abi = general_helpers.load_abi(constants.path_erc20_abi)
uniswap_v3_pair_abi = general_helpers.load_abi(constants.path_uniswap_v3_pair_abi)

# Parse all events from Uniswap v3 transaction
events = parse_uni_v3_events.parse_all_v3_events(logs, erc20_abi, uniswap_v3_pair_abi)
```

Acknowledgements

I thank the teams at (TrueBlocks Team, 2022) and (Erigon Team, 2022) for their support on node operation and indexing, as well as community members of Flashbots and Uniswap for their help with understanding the DeFi protocols.

References

- Adams, H., Zinsmeister, N., & Robinson, D. (2020). *Uniswap v2 core*. Tech. rep., Uniswap.
- Buterin, V. (2013). *A next-generation smart contract and decentralized application platform*. White paper. <https://ethereum.org/en/whitepaper/>
- Erigon Team. (2022). *Erigon*. <https://github.com/ledgerwatch/erigon>
- TrueBlocks Team. (2022). *TrueBlocks: Lightweight indexing for any EVM-based blockchain*. <https://trueblocks.io/>
- Wood, G. (2014). *Ethereum: A secure decentralised generalised transaction ledger*. Yellow paper. <http://gavwood.com/Paper.pdf>